Fiscal Year:	FY 2007	**Task Last Updated:**	FY 02/13/2008
PI Name:	Shen, Hayley H Ph.D.		
Project Title:	Constitutive Relation in Transitional Granular Flows		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	02/13/2002	**End Date:**	12/31/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hanes, Daniel (University of Florida/Hanes and Associates)		
Grant/Contract No.:	NAG3-2717		
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	The behavior of granular materials spans the whole range from an elastic solid to a non-linear viscous fluid. The constitutive relation depends not only on the material composition, but also on the deformation and the rate of deformation. The key reason for this behavior is that granular materials are not only defined by the properties of the individual grains, but also by the structure of the aggregate. In a very dilute granular flow, the structure of the aggregate is simply defined by a random velocity distribution. As the concentration increases, the internal structure becomes complex. The internal structure evolves dynamically with the macroscopic deformation and is an integral part of the constitutive relation. The complexity and importance of the internal structure increase with the concentration. In a gravity field, the weight of a granular material naturally compacts the material to produce a concentration gradient. To study the constitutive behavior of granular materials, microgravity experiments must be performed to construct a		

Task Description:	physical basis for transitional constitutive relations. For long-term exploration on the Moon and the Mars, rheological properties of the surface granular materials require a better understanding. Moreover, many industrial problems on Earth deal with processing granular materials. Improving fundamental understanding of granular materials will also impact on industrial processes. This project combines both numerical and physical experiments to study the transition of granular flows from solid to gas phases. Using these experiments we have obtained data to quantify the internal resistance of a deforming granular material, to define and measure the internal structures, and to formulate theories to describe the transitional behavior. This project includes collaborators: Dan Hanes, Hanes and Associates, and James T. Jenkins, Cornell University. This collaboration has been extremely beneficial.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Improving the fundemental understanding of the rheological properties of granular materials will impact industrial processes such as pharmaceutical powder processing and design of construction (i.e. earth moving) equipment. It will improve engineering design and maintenance of structures to withstand landslides, snow and debris avalanches. Provide necessary knowledge to help NASA prepare for upgrading new launch fuel facility.
Task Progress:	1. Built 3-d code for numerous applications: angle of repose, incline flow, hourglass, penetration tests, simple shear flows. 2. Included electrostatic forces, cluster particles with and without overlaps. 3. Finished building two annular shear cells, one 8-in diameter and one 24-in diameter. They will be used to test scaling effects. The co-existence of two annular shear cells is unique in the research world. 4. Introduced force chain measures to granular flows. Ready for next step of completely mapping the transitional behavior from dilute to dense phase. 5. Obtained physical data from collaborator Dan Hanes and compared with simulation data. 6. Completed a regime diagram to classify the transition between phases. A biology paper is published from a collaboration initiated by the PI when attending the Microgravity training workshop in Cleveland in 2004. This paper is success story between an engineer and a group of biologists. It is also a success story for having the microgravity workshop that produces interactions between many researchers in different disciplines. Two papers are under preparation: - Ji, S. and Shen, H.H. (2007) Shape factor on bulk friction of granular materials, under preparation. - Ji. S., Hanes, D.M. and Shen, H.H. (2007) Testing Discrete Element Simulations of Granular Materials using Annular Shear Cell Experiments, under preparation.
Bibliography Type:	Description: (Last Updated: 02/13/2008)
Articles in Peer-reviewed Journals	Ji S, Shen HH. "Internal parameters and regime map for soft polydispersed granular materials." Journal of Rheology 2008 Jan-Feb;52(1):87-103. http://dx.doi.org/10.1122/1.2807441 , Jan-2008
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Papers from Meeting Proceedings	Ji S, Shen HH. "Control Parameter for Phase Transition of Granular Materials." 15th USNCTAM06, Boulder, Colorado, June 25-30, 2006. Proceedings 15th USNCTAM06, Boulder, Colorado, June 25-30, 2006. , Jun-2006
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Papers from Meeting Proceedings	Shen HH, Jenkins JT, Ji S. "Geometrically Constrained Flow Behavior with Uniform Disks." The 2005 Joint ASME/ASCE/SES Conference on Mechanics and Materials, Baton Rouge, Louisiana, June 1-3, 2005. Proc. McMat2005, The 2005 Joint ASME/ASCE/SES Conference on Mechanics and Materials, Baton Rouge, Louisiana, June 1-3, 2005, Paper No. 624. , Jun-2005
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